Kyra Gunluk

Contact

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Education

Georgia Institute of Technology *Atlanta, GA*
PhD in Algorithms, Combinatorics, and Optimization *Fall 2024-Spring 2029 (Expected)*
Advised by Professor Santosh Vempala, based in College of Computing

Cornell University *Ithaca, NY*
B.S. in Computer Science, Magna Cum Laude *Fall 2020-Spring 2024*
Minors in Operations Research, Mathematics, and Game Design

Research Experience

Visiting Researcher Columbia University *Summer 2026*
Conducting research on iterated central path methods and their complexity for reaching polytope vertices, with Professor Bento Natura.

Graduate Research Assistant Georgia Institute of Technology *Fall 2024 - Current*
Conducting research on topics in Approximation Algorithms with Professor Santosh Vempala. We focus on fundamental objects in geometric and combinatorial optimization, including polytope volumes and graph sparsest cuts.

NSF DREU Research Intern UIUC, Champaign, IL *Summer 2023*
Selected as an on site participant for 10 weeks in Urbana-Champaign, working with Professor Ruta Mehta. Conducted research on Algorithmic Game Theory, and developed approximation algorithms for fair allocation of indivisible items among agents.

Undergraduate Research at Cornell Cornell University, Ithaca, NY *Fall 2022, Spring 2023*
Worked on a research project on Fourier Bounds for Parity Decision Trees (PTDs) with Professor Eshan Chattopadhyay. The goal of this project is to improve current bounds on the complexity of Fourier expansions of boolean functions on binary strings.

Written Work

Deterministic Volume Estimation of Truncated Hypercubes
Kyra Gunluk

Approximating MMS and (symmetric) APS under Cardinality Constraints: Goods, Bads, and the Best-of-Both-Worlds
Arjun Aggarwal, Kyra Gunluk, Ruta Mehta

Academic Presentations

MMS and APS Under Heterogeneous Cardinality Constraints (Poster Presentation)
- Workshop in Honor of Mihalis Yannakakis, Columbia University *August 2023*
- EaGL Theory of Computation Workshop, Univeristy of Rochester *September 2023*

Deterministically Approximating Volume of a Hypercube Intersected with Halfspaces
- Talk at the ACO Student Seminar, Georgia Tech *November 2024*
- Poster and flash talk at ACORN 2025, FriezeFest, Carnegie Mellon University *October 2025*
- Flash talk at the ARC Colloquium, Georgia Tech *August 2026*

Teaching Experience

Georgia Tech, Atlanta, GA, CS Course Staff *Spring 2026*
TA for CS 4540: Advanced Algorithms. Held hour long office hours to assist students in problem solving, graded all homeworks and exams.

Cornell University, Ithaca, NY, CS Course Staff *Spring 2022, Fall 2022*
TA for CS 2800: Discrete Structures. Taught hour long discussion sessions twice a week, graded weekly homeworks and three exams, held office hours.

Cornell University, Ithaca, NY, CS Course Staff *Spring 2023*
Teaching Assistant for CS 4820: Introduction to Analysis of Algorithms. Held hour long office hours to assist students in problem solving, graded all homeworks and exams.

Cornell University, Ithaca, NY, Engineering Learning Initiatives *Fall 2021*
Academic Excellence Workshop Facilitator for Math 2930: Differential Equations for Engineers. Created weekly lesson plans, lecture slides, and worksheets for students; taught weekly 2-hour sessions, during which I presented review slideshows, assisted students and provided full solutions.
Georgia Tech, Atlanta, GA, ISyE mentor *Fall 2025*
Graduate Mentor for CSP (Community Service Project) with ISyE's Center for Academics, Success, and Engagement. Mentoring undergrad teams through applied projects with industry partners

Educational Outreach

EdEquity, Denver, CO, *College-In-High School* (remote) *Fall 2023*
TA for ENGRI 1101: Engineering Applications of ORIE for High School Students. Led coding lab sections twice a week, held office hours, and graded homeworks and exams.

Discovery Program, New York, NY, Stuyvesant HS *Summer 2022*
Teaching Assistant for mathematics, Algebra 1. Provided guidance for problem solving and graded homework assignments and exams.

STAR Learning Center, New York, NY, Tutor *March 2019 – July 2019*
STEM tutor for underprivileged students. Created individual lessons, reported progress afterwards.

Applied Projects

Action IQ Research Internship, New York, NY, Intern *Summer 2019*
Research project on the relationship between weather conditions and shopping trends at various Intermix stores. Worked with a partner using python to create graphs and analyze the data to relate customer shopping patterns to weather data.

Salvage Computer Game, Cornell University, Designer *Spring 2022*
Worked with a team to develop a deep sea diving exploration video game. Character Design of the diver and the monsters, creating animation cycles for different movements using Adobe Photoshop.

Lunar Lasso Mobile Game, Cornell University, Designer *Spring 2023*
Worked with a team to develop a space cowboy obstacle game. Character Design for all game assets, creating animation cycles for different movements using Adobe Photoshop.

Hospitality Hackathon, Cornell University, Researcher *2021*
Worked with a team to form theoretical solutions to current Hilton technology and staffing issues and presented our ideas.

Honors

ARC-ACO Student Fellowship *2026*
ACO Fellowship *2024*
Member of Google CSRMP (Computer Science Research Mentorship Program) *2022*
Participant in the New Horizons in Theoretical Computer Science Summer School *2023*
Member of Study of Exceptional Talent, Johns Hopkins Center for Talented Youth *2015*

Relevant Skills

Languages: English - Native, Spanish - Intermediate
Programming: Python, Java, C, HTML, OCaml, MATLAB, OR-Tools, CVXPY, LaTeX
Design: Word, Excel, Adobe Photoshop, Procreate

Professional Memberships

Society of Women Engineers
Women in Computing at Cornell